Matthew Gardner

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SUMMARY

I'm a **Senior Data Engineer** with a solid background in data engineering and hands-on experience in building scalable **data pipelines** and working with **big data technologies** across **cloud platforms**.
I also have a strong foundation in machine learning and predictive analytics, allowing me to bridge the gap between data engineering and data science.
My goal is to deliver end-to-end data solutions that not only move and store data efficiently but also uncover insights that drive real business impact.

EDUCATION

BS in Computer Science | Carnegie Mellon University **2008 - 2012**

SKILLS

Programming: Python, SQL, PySpark, Scala, Java, Shell Scripting, Javascript

Cloud Platforms:

- **AWS:** S3, Lambda, Glue, EMR, Redshift, Athena, DynamoDB, Step Functions
- **Azure:** Data Factory, Databricks, Azure SQL
- **GCP:** BigQuery

Data Engineering & Pipeline: ETL/ELT, dbt, Apache Nifi, Apache Kafka, Spark, Big Data(Hadoop, Spark), Apache Kafka, Amazon Kinesis, Warehousing(Snowflake, BigQuery, Redshift), Lakehouse (Delta Lake, Iceberg), Modeling, CDC

Databases: PostgreSQL, MySQL, MongoDB, SQL Server, DynamoDB, Aurora, SQL/NoSQL, Oracle

Infrastructure & DevOps: Docker, Kubernetes, Terraform, GitHub, Jenkins, CI/CD

Monitoring & Orchestration: Prometheus, Grafana, Datadog, Apache Airflow, CloudWatch

Machine Learning & Analytics: Scikit-learn, NumPy, Pandas, Unity Catalog, Databricks, MLOps, Matplotlib

Analytics & BI: Power BI, Tableau, Excel, Looker, Sigma

Integration & Data Formats: REST APIs, Webhooks, SFTP, JSON, Parquet, Avro

Soft Skills & Leadership: Cross-Functional Collaboration, Stakeholder Communication, Agile Methodologies, Mentoring

EXPERIENCE

Senior Data Engineer | Optum 02/2021 - Current

- Partnered cross-functionally with business and clinical teams to design and launch **data platforms** aligned with AI-driven initiatives, improving clinical and operational decision-making.
- Designed and built real-time **CDC** and **data pipelines** with **Apache Kafka** (MSK), **Spark**, and **Hadoop**. I optimized high-velocity stream processing, enabling businesses to make real-time decisions without delays.
- Redesigned data infrastructure by implementing a **Databricks Lakehouse** using **Delta Lake** and **PySpark** with **Unity Catalog**, bringing together structured and semi-structured data into a single, unified analytics environment. This upgrade streamlined workflows for both analytics and machine learning, improving data processing efficiency and cutting model deployment time, making insights more accessible and actionable.
- Moved event-driven data processing to **AWS Lambda**, eliminating the need for traditional server infrastructure and cutting operational costs. This made it easier to scale up or down based on demand, ensuring smooth performance even with fluctuating workloads. Also improved cloud resource provisioning using **Terraform**.
- Redesigned data architecture to support complex workflows, using scalable models like star, snowflake, and dimensional schemas. This upgrade improved query efficiency, helping teams to quickly access insights and make more data-driven decisions immediately.
- Developed **ETL workflows** using **Azure Data Factory, Dagster, dbt** and **Apache Airflow** to ingest and transform data from diverse sources including **PostgreSQL** and accelerate processing across business teams. I also developed **ETL pipelines** for ingesting and transforming data from diverse sources into **Redshift**, and used **AWS Glue** and **Lambda** for automation.
- Developed and deployed **ML models** on AWS SageMaker and GCP Vertex AI. After several iterations, improved model accuracy. Also streamlined cloud setup to cut training time in half.
- Worked with a team to build a data lake on **GCP**, combining **BigQuery** and Cloud Storage to centralize vast datasets for easier access.
- Introduced automated data quality checks with **Great Expectations** and **Delta Lake constraints**, improving data accuracy to around 99% across the **ETL pipeline**. This automation eliminated tedious manual validation, saving the team significant time per week, helping engineers to focus on deeper analysis rather than routine data-cleaning tasks.
- Optimized data storage strategy for high-traffic platforms using **Redshift** and **Snowflake** to handle massive datasets efficiently. This reduced dashboard and ad-hoc query latency by 30-50%.
- Worked with business and clinical stakeholders to design **Looker** dashboards and **KPIs**, enabling self-service analytics and informed decision-making.
- Mentored and supported junior data engineers, helping them sharpen their technical skills and navigate complex projects. I regularly provided hands-on training, troubleshooting, and career guidance, fostering a collaborative and high-performing team environment.

- Integrated ML models into cloud-based systems to build scalable **MLOps** pipelines for real-time claims fraud detection.
- Monitored the system using **Prometheus**, **Grafana**, and **AWS CloudWatch**.

Data Engineer | Accenture 08/2017 – 01/2021

- Simplified **ETL workflows** by automating key tasks with **Apache Airflow**, cutting down manual processes and freeing up valuable engineering time.
- Improved reporting systems by integrating real-time data feeds into **Tableau** and other **BI platforms** to make reports faster and more accurate. This upgrade helped teams to access insights shortly and businesses make immediate decisions.
- Set up real-time data monitoring with **AWS CloudWatch** and **Logstash**, allowing early detection of bottle necks in ETL processes. I optimized workflows, ensuring smoother data movement and faster processing times, by continuously tracking performance metrics.
- Processed and refined large datasets using Python (**Pandas**, **NumPy**) and SQL, ensuring data accuracy for reporting and in-depth exploratory analysis.
- Built basic predictive models using **Scikit-learn**, such as logistic regression and decision trees, to support churn prediction and customer segmentation projects.
- Created interactive data visualizations and dashboards using **Tableau** to communicate findings to business stakeholders and support data-driven decision-making.
- Conducted hypothesis testing and statistical analysis to evaluate the impact of marketing campaigns and product changes, improving targeting strategies.
- Automated repetitive data collection and reporting tasks with Python scripts, reducing manual work and increasing operational efficiency.
- Collaborated with data engineering and business teams to define **KPIs** and ensure accurate tracking and reporting of key metrics.
- Worked closely with data scientists, analysts, and product teams to ensure the accurate delivery of data insights.
- Participated in team code reviews and shared knowledge of Python libraries and best practices in collaborative analysis environments.

Data Analyst/Engineer | Fannie Mae 02/2013 - 07/2017

- Performed **risk analysis** and **portfolio segmentation** using SAS, SQL, and Excel, supporting under writing decisions and reducing exposure to high-risk borrowers by 18%.
- Developed and deployed regression models in **R** and **Python** to forecast loan default probabilities, improving credit policy effectiveness.
- Conducted customer behavior analysis to identify cross-sell and up-sell opportunities, leading to a 12% increase in product adoption.
- Created automated dashboards using **Tableau** and **QlikView**, streamlining reporting workflows for risk and fraud monitoring teams.
- Worked with large-scale transactional data to identify trends and anomalies, supporting the compliance team in detecting potential fraudulent activity.

- Collaborated with product managers to run **A/B tests** and evaluate the impact of digital features on user engagement and retention.
- Partnered with engineering teams to move analytics from ad-hoc reporting to reusable **SQL scripts** and scheduled ETL pipelines.
- Presented insights and visualizations to non-technical stakeholders, helping influence product and policy changes with data-driven evidence.